

1940's ==> write the code / program

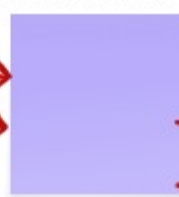
addition ==> 010101001001

subtraction ==> 010111101010

.....

.....

CPU / Processor



0's and 1's

output

output

The program that is written using 0's and 1's ==> Machine Level Language(MLL)

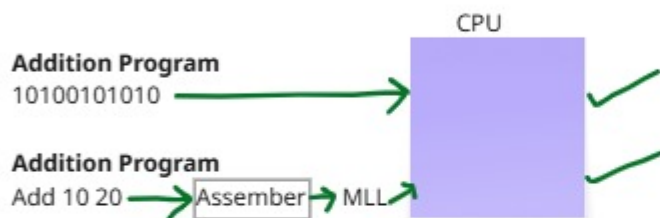
Problem ==> It is difficult to remember the program written using 0's and 1's

1950's ==> **Assemble Level Language** ↓ neumonics

addition ==> ~~010101001001~~ ==> Add 10 20

subtraction ==> ~~010111101010~~ ==> Sub 10 20

ALL is a language that is written using neumonics

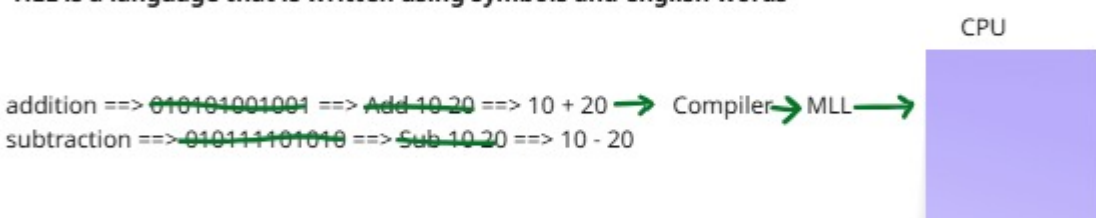


1960's ==> **High Level Language**

addition ==> ~~010101001001~~ ==> ~~Add 10 20~~ ==> 10 + 20

subtraction ==> ~~010111101010~~ ==> ~~Sub 10 20~~ ==> 10 - 20

HLL is a language that is written using symbols and english words



BCPL language ==> 1962 ==> Martin Richards

==> This language was unstructured language

==> This language was occupying more memory

unstructured means ==> we cannot reuse the program

```
-----  
-----  ADD  
-----  
  
-----  
-----  SUB  
-----  
  
-----  
-----  MUL  
-----  
  
-----  
-----  ADD  
-----
```

Ken Thompson ==> 1969 ==> B language

BCPL ==> More Memory, Unstructured

B ==> Less Memory, Unstructured

Dennis Ritchie ==> 1972 ==> C language

Less Memory, Structured

int age = 50 ==> Global Variables

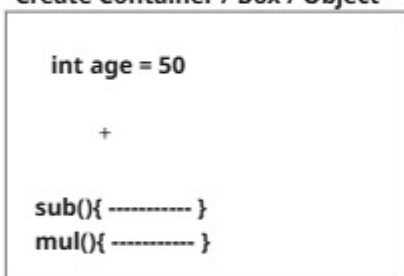
```
add()  
sub()  
div()  
mul()  
add()
```

```
add() { ----- }  sub() { ----- }  Mul() { ----- }  div() { ----- }
```

1982 ==> Bjarne Stroustrup ==> C++

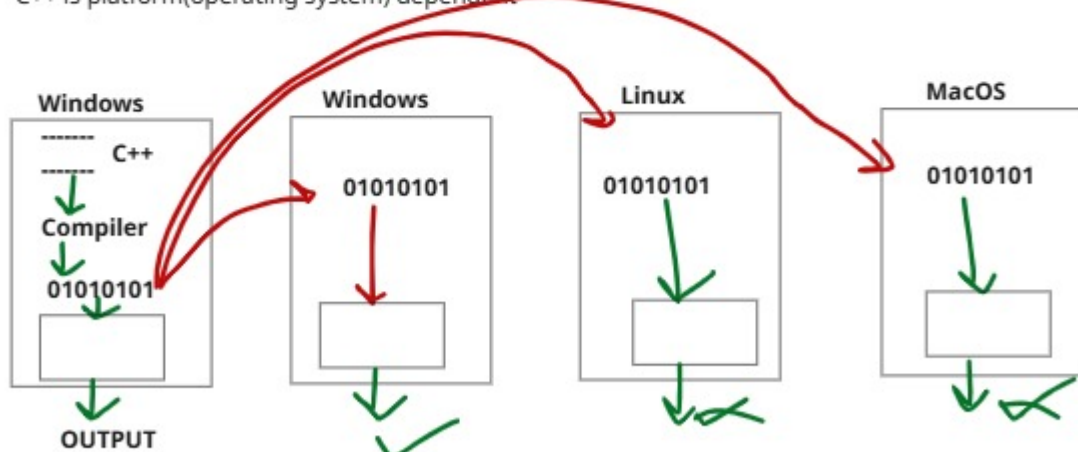
Inveneted a language that solves the problem of C language

Create Container / Box / Object



Problem with C++

C++ is platform(operating system) dependent



James Gosling ==> **JAVA** ==> Language ==> Less memory, Objects Concept, Platform Independent,

JAVA ==> 1992 ==> unofficially

JAVA ==> 1996 ==> officially